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Evaluating mobile robot navigation behavior in flexible assembly systems through digital twin and real-world experiments

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Abstract

Although digital twins are increasingly used for pre-deployment testing, their reliability as predictive tools remains understudied due to the lack of established validation frameworks. This paper presents a systematic methodology for validating the predictive fidelity of physics-based digital twins in robotic navigation tasks, addressing a critical gap in sim-to-real transferability for industrial mobile manipulators. We propose a novel evaluation approach combining (1) multi-metric comparison (localization accuracy, path consistency, goal accuracy, and navigation performance) between real-world and simulated navigation experiments, and (2) an uncertainty quantification method to establish confidence intervals for digital twin predictions. Using an *NVIDIA Isaac Sim* model of an omnidirectional mobile manipulator and digitally reconstructed production environments, we conduct 50 real-world and 50 digital twin experiments across five industrial scenarios. The results show a mean Hausdorff distance of 0.195 m between real and simulated paths, localization RMSE differences of 0.005 m, and a path prediction accuracy of ±0.229 m (95% CI). The findings contribute to robotic navigation by addressing key challenges in using digital twins for real-world applications, reducing the sim-to-real gap, and enhancing the reliable deployment of mobile manipulators in flexible assembly systems.

Keywords Mobile robotics, Digital twin, Sim-to-real transfer, Navigation, Localization

1 Introduction

1.1 Challenges in modern industries

The aging population and the growing shortage of skilled workers present pressing challenges for modern industries, driving the adoption of automation in assembly systems to maintain productivity and competitiveness [1–4]. Concurrently, evolving market demands for greater product variety, mass customization, shorter production cycles, and increasing price pressure require automation systems that are not only efficient but also highly flexible and reconfigurable [5]. Traditional automation systems, originally

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designed for repetitive tasks in static environments, are struggling to meet the growing demands of modern manufacturing [6]. In response, mobile manipulators are increasingly integrated into flexible assembly systems, enabling autonomous operations in dynamic production environments. However, their successful deployment depends on precise localization and reliable navigation to ensure safe and efficient movement. Any changes to such systems, whether due to updated workflows, reconfigured layouts, modifications in the robotic system, or interactions with human workers, can introduce uncertainty in the execution of robotic processes [7–9]. Without proper investigation, these uncertainties can lead to process failures, resulting in costly downtime, and ultimately impact overall productivity and profitability.

1.2 Research gap and objectives

Physics-based digital twins offer a promising solution to address these challenges. A digital twin is a real-time, bidirectionally connected virtual representation of a physical system that enables continuous synchronization and interaction [10]. By maintaining a live link between the physical and digital domains, digital twins support the simulation, evaluation, and optimization of system behavior prior to and during real-world operation. This capability reduces uncertainty and enhances confidence in process outcomes, enabling companies to reduce risks associated with system changes. Although digital twins hold great potential, their practical reliability for specific applications, such as predicting robotic navigation behavior, has not been sufficiently explored. In particular, standardized metrics for comparing real and simulated navigation behavior, as well as methods for quantifying path uncertainty in digital twin predictions, are still lacking. This gap limits the confidence industries can place in robotic digital twins as predictive tools, especially when evaluating navigation path selection and execution after system changes. By comparing the localization and navigation behavior of a real robot and its digital counterpart in *NVIDIA Isaac Sim* across various scenarios, this paper evaluates the accuracy, robustness, and practical applicability of robotic digital twins for navigation. Furthermore, this study quantifies the uncertainty of navigation predictions in the digital twin, assessing the transferability of the digital robot's behavior to real-world navigation processes in flexible assembly environments.

1.3 Structure of the paper

This paper is organized as follows. Section 2 reviews related work in the field of digital twins and robotics. Section 3 details the evaluation setup, covering scenarios, hardware, the digital twin, as well as communication and software architectures. Section 4 presents the methodology, covering experiment preparation and execution, evaluation metrics, and the quantification of navigation uncertainty. Section 5 discusses the results of the experiments. Finally, Sect. 6 concludes the paper and outlines directions for future work.

2 Related work

The dynamic nature of modern manufacturing environments demands systems capable of rapid reconfiguration to accommodate varying production requirements [11, 12]. Achieving this level of flexibility is made possible through the integration of mobile manipulators, which improve resource utilization and task flexibility by facilitating the dynamic adaptation of workflows [13–16]. For mobile manipulators operating in

modern manufacturing environments, localization and navigation capabilities are fundamental, allowing them to determine their pose and navigate to key locations, such as assembly or logistics stations [17, 18]. To simulate and test robotic behavior prior to implementation in real-world scenarios, digital twins, virtual replicas of physical systems, have emerged as a powerful tool [19–22]. While prior work demonstrates their utility in the design-phase of navigation systems (e.g., path planning [23] and semantic environment modeling [24]), these studies focus on functional correctness rather than predictive fidelity, the ability to forecast real-world behavior. Further research examines the use of digital twins to compare navigation algorithms for mobile robots [25], but they lack a systematic evaluation of sim-to-real transferability, a limitation that our research targets. Research efforts continue to address the gap between simulated and real-world robot performance, with one study exploring the optimization of simulation parameters to enhance the transferability of learned behaviors from simulation to real-world robots [26]. Another study benchmarks the sim-to-real gap in agile manufacturing by employing high-fidelity digital twins and focuses on tasks like bin-picking and assembly [27]. While primarily centered on manipulation tasks, these studies do not analyze robotic navigation behavior, a critical aspect of our research. The extent to which simulation performance predicts real-world outcomes for visual navigation tasks is explored in [28]. However, it does not consider industrial-like settings and omits the use of mobile manipulators for automation in manufacturing environments. Another study introduces a digital twin framework for automating transportation with AGVs in industrial settings, focusing primarily on synchronizing mission times between digital and physical systems [29]. Yet, it does not examine navigation discrepancies, which are essential for evaluating the fidelity of digital twins in replicating real-world navigation behavior. Recent research has introduced various metrics to quantify the sim-to-real gap in robotics. One study proposes the Instance Performance Difference metric, designed to measure discrepancies in perception tasks by comparing performance on synthetic versus real images, specifically in camera simulations [30]. Another work introduces the Velocity Estimation Performance Difference metric, focusing on the accuracy of GPS and IMU sensor simulations by evaluating velocity estimation discrepancies between simulated and real-world data [31]. While these studies contribute valuable insights into the sim-to-real gap, they primarily focus on perception and sensor simulation rather than the behavior of the entire robotic system.

Despite these advancements, bridging the sim-to-real gap for reliable navigation remains a challenge. Our work addresses this gap by introducing a multi-metric validation protocol for industrial mobile manipulators, quantifying navigation uncertainty via confidence intervals, and demonstrating applicability in industrial-like scenarios.

3 Evaluation setup

3.1 Experimental scenarios

To address the need for a comprehensive uncertainty quantification method, five different Navigation scenarios are conducted in both the real-world and the digital twin, evaluating localization accuracy, goal accuracy, path consistency, and navigation performance. To ensure reliability and statistical significance, each of the navigation scenarios is executed 10 times, resulting in a total of 50 real-world and 50 digital twin Navigation experiments. In these experiments, the robot navigates from predefined start poses to

goal poses within a controlled, 44 m² reconfigurable production environment, featuring assembly and logistics stations, a collaborative robot, and a diverse set of obstacles. These elements introduce challenges such as narrow passages and constrained maneuverability, replicating the complexities of modern production environments. Depending on the scenario, the path between start and goal pose is either unobstructed or obstructed by obstacles, which may be known (included in the map used for localization) or unknown (absent from the map used for localization, yet considered by real-time sensor inputs). Figure 1 displays the occupancy grid maps for each scenario.

3.2 Hardware setup

The experiments utilize the *RB-Kairos* mobile manipulator platform, featuring an omnidirectional wheelbase for maneuvering in confined spaces and an *UR10* robotic arm for manipulation tasks [32, 33]. The robot's sensor suite includes an Inertial Measurement Unit (IMU), Hall effect sensors, two *SICK S300* laser scanners, and an *Ouster OS1-128* 3D LiDAR. To accurately track the real-world robot's pose, the photogrammetry system *OptiTrack* with 38 *Prime*^{x}*41* cameras is utilized. This motion capture system provides millimeter-level accuracy, allowing for detailed comparisons with the digital twin's performance [34]. For pose tracking, the physical robot is equipped with spherical markers placed on the robot's base aligning the geometrical marker center with the known frame *base_footprint* of the robot. Two 3D-printed anchors, equipped with markers, are utilized to define the start and goal poses of the Navigation tasks, ensuring repeatability of navigation runs through the scenarios. Strategically positioned 5 G radio dots form a 5 G Radio Access Network (RAN) that covers the shop floor and connects an application server and the mobile manipulator system.

3.3 Digital twin setup

The system presented in this study is part of a broader digital twin framework. During the experiments, the digital counterpart functions as a digital shadow, receiving data from the physical robot without influencing its behavior. Although the setup described below aligns more closely with the concept of a digital shadow, the system is designed to support bidirectional data exchange. Therefore, the term digital twin is used throughout this study to reflect its intended design. The digital twin of the real-world system is created using *NVIDIA Isaac Sim* [35]. *NVIDIA Isaac Sim* leverages the *PhysX* engine

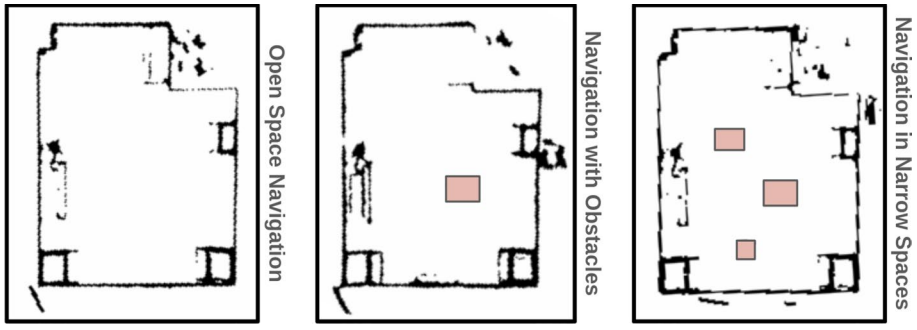


Fig. 1 Occupancy grid maps generated in the real-world setup for the scenarios 'Open Space Navigation 01' (left), 'Navigation with Obstacles 02/03' (middle), and 'Navigation in Narrow Spaces 04/05' (right). Obstacles in the 'Navigation with Obstacles' and 'Navigation in Narrow Spaces' scenarios may be either unknown (scenarios 02 and 04) or known (scenarios 03 and 05). Obstacles are highlighted for improved visibility

to model robot motion dynamics and incorporates ray tracing technology for realistic sensor simulation. The digital twin integrates a robot model, sensor models, action graphs, and an up-to-date environmental model to ensure that both the digital robot and the environment closely mirror their real-world counterparts. The robot model is created in *NVIDIA Isaac Sim* using the built-in URDF importer extension, aligning the virtual model's dimensions, joint configurations, physics parameters, and sensor placements with those of the physical robot. Mecanum wheels are modeled with sphere colliders approximating roller interactions, while joint parameters are fine-tuned through experimentation for realistic motion dynamics. Figure 2 shows a side-by-side comparison of the physical robot and its digital counterpart. The sensor models, including the 3D LiDAR and laser scanners, are parameterized using configuration files from the *NVIDIA Isaac Sim* sensor extension, generating outputs directly comparable to those of the physical sensors. Action graphs serve as the backbone for managing the way robots and sensors behave. The action graph for base actuation governs the wheel joint control, enabling the robot to move. The sensor action graphs manage the continuous flow of data from the simulated sensors by publishing sensor readings. The digital replica of the physical environment is constructed using a systematic post-processing workflow,



Fig. 2 Side-by-side comparison of the physical robot and its digital counterpart, highlighting the design of the mecanum wheels in the digital model

converting point cloud measurements from the real-world mapping phase into a 3D mesh, as detailed in Sect. 4.1.

3.4 Communication and software setup

The communication and software framework for the robotic system is based on the *Robot Operating System* (ROS2), which provides the middleware and essential tools for communication and coordination between the software components deployed across the robot's on-board computer and the application server [36]. Wireless communication is established using a 5 G network and the *zenoh* middleware, enabling high-speed, low-latency data transmission, which is crucial for dynamic and responsive robot operations [37]. A simplified illustration of the communication and software setup for the real robot is shown in Fig. 3. Communication between the digital robot and the ROS2 software stack is facilitated through the *ROS2 bridge extension*, allowing the digital robot to publish and subscribe to ROS2 topics, services, and actions. The ROS2 bridge enables both the digital and real-world robots to operate on a unified software stack, ensuring consistent control execution and sensor data processing across the two systems.

Robot-side Software. On the physical robot, motion execution is managed by the robot control software, which interprets high-level commands and translates them into motor commands for the mobile base. Sensor drivers enable the acquisition of data from various sensors, including laser scanners, 3D LiDAR, IMU, and rotary encoders. Accordingly, the robot-side software closely parallels the functionality managed by the action graphs in the digital twin. To minimize bandwidth usage, data filtering and compression techniques are employed, particularly for the transmission of point cloud data between the robot and the application server.

Application Server-side Software. The application server acts as the central intelligence hub, hosting *NVIDIA Isaac Sim* (version 4.2.0), ROS2 modules for both the physical robot and its digital counterpart, and a client for streaming motion capture data from the *OptiTrack* system. Simultaneous Localization and Mapping (SLAM) is performed

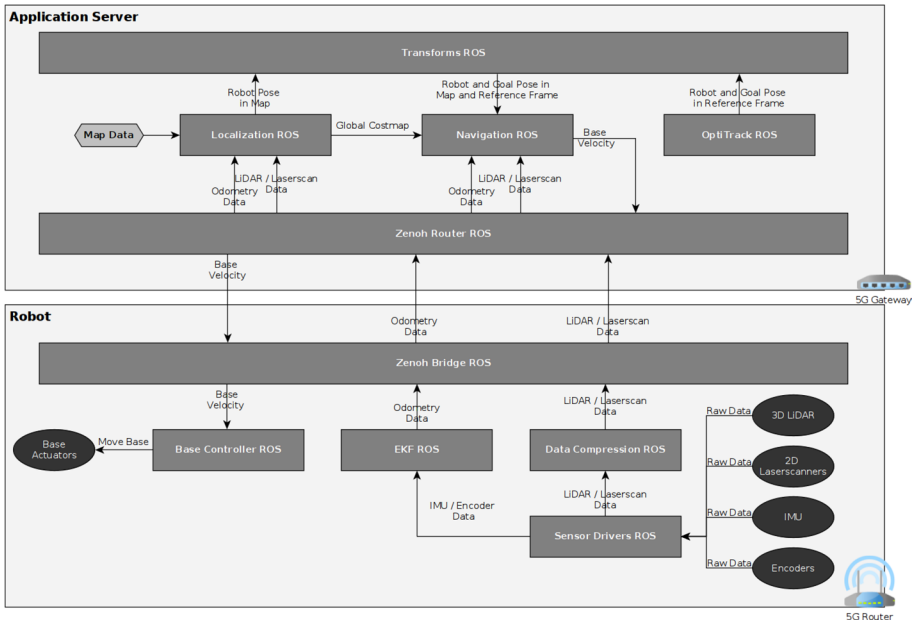


Fig. 3 Communication and software setup for the real-world experiments

using the *Google Cartographer* algorithm, which relies on sensor inputs from the 3D LiDAR and odometry data derived from an Extended Kalman Filter (EKF) that fuses IMU and encoder measurements [38]. Navigation is managed by the *Nav2* framework, utilizing inputs from the SLAM system, spatial sensor data from 3D LiDAR and laser scanners, and odometry information [39]. Pathfinding is conducted using the A^* algorithm, while the *DWB Local Controller* handles local motion control [40]. Both local and global costmaps are employed, incorporating static, obstacle, denoise, and inflation layers. Goal completion criteria are configured with an absolute tolerance of $0.1\,\mathrm{m}$ and 6°. For further details on the software and communication framework, see our previous work [41].

4 Methodology

4.1 Experiment preparation

The experiment preparation starts with mapping, where the robot is manually guided through the physical environment. Sensor data, trajectory information, and motion capture reference data are recorded in bag files. Three datasets corresponding to the scenarios (see Fig. 1) are collected and post-processed for digital experiments. The post-processing workflow includes:

- *Bag Extraction*: Time-stamped point cloud data and transformations from motion capture and SLAM systems are extracted from the recorded mapping bag files.
- *Pose Interpolation*: Interpolation is applied to the transformations, estimating the robot's pose at the timestamps of the point clouds.
- *Point Cloud Transformation*: Point clouds are transformed from the LiDAR frame to the global motion capture reference frame.
- *Point Cloud Filtering*: The transformed point cloud is filtered to remove outliers and ground points and downsampled using a voxel-based approach.
- *3D Mesh Generation*: A 3D mesh representation is generated by assigning voxel cubes to the points, combining them into a 3D mesh.
- *Data Saving*: The resulting 3D mesh, along with the robot's trajectory information are saved.

To map the digital environment, *NVIDIA Isaac Sim* and the ROS2 software stack, identical to that used in the real-world setup, are launched. The digital twin is initialized by setting up the world, timeline, and physics scene, followed by the creation of the ground plane. The 3D environmental mesh, derived from the real-world mapping phase, and the robot model are imported into the scene and aligned to match their real-world poses. Once initialized, the digital twin progresses in discrete steps, rendering the environment and performing physics computations while the robot is manually guided through the environment for mapping. Figure 4 shows a side-by-side comparison of the real-world and digital environments and their corresponding SLAM-based maps.

4.2 Experiment execution

Real-world navigation experiments begin by launching the ROS2 software stack, including the navigation framework and the SLAM algorithm in pure localization mode, using the scenario-specific map file. Motion capture anchors are placed to define the start and goal poses. Before execution, the robot is localized using SLAM within the map and

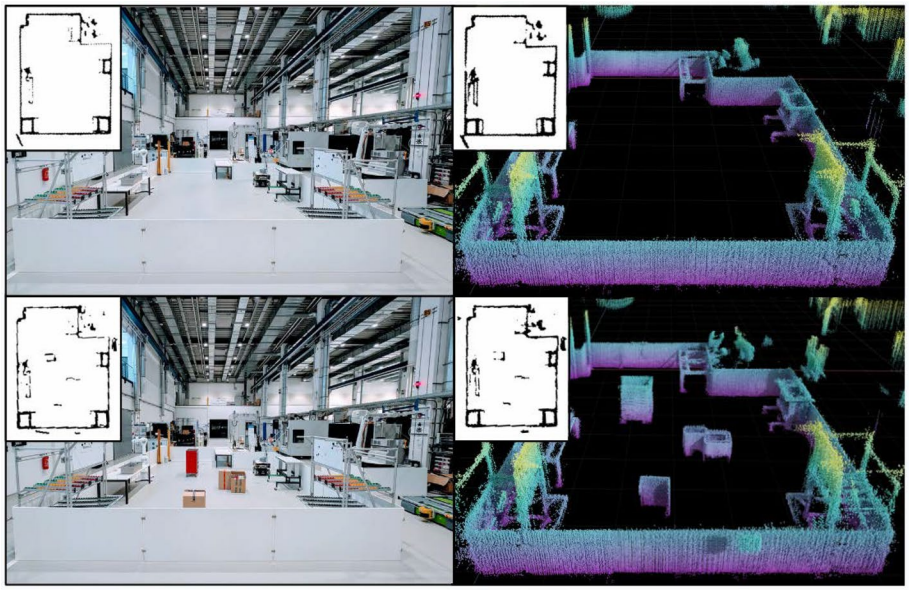


Fig. 4 'Open Space Navigation' (top) and 'Navigation in Narrow Spaces' (bottom) scenarios, showing the real-world setup (left), digital twin replication (right), and their corresponding SLAM-based maps

aligned with the start pose anchor. The goal pose is then provided to the *Nav2 BasicNavigator* interface, which directs the robot toward the goal while avoiding obstacles. During each trial, a bag file with key data, including motion capture and SLAM trajectories, is recorded for evaluation. Each experiment is then replicated in the digital twin. The digital twin is initialized following the workflow in Sect. 4.1, loading the 3D environment and robot model with poses aligned to real-world motion capture data. After launching the ROS2 software stack, the goal pose is passed to the *Nav2 BasicNavigator* interface, which generates control commands. These commands are transmitted via the ROS2 bridge to the action graph, actuating the robot toward the goal. As in real-world trials, a bag file is recorded and relevant evaluation data is extracted.

4.3 Evaluation metrics

To evaluate the performance of the robot's behavior in both the real world and its digital twin, we employ quantitative metrics for localization accuracy, goal accuracy, path consistency, and overall navigation performance. These metrics are essential for assessing how closely the digital twin replicates the real-world behavior of the robot, which is crucial for validating the predictive fidelity of the digital twin. By systematically comparing these metrics across multiple real-world and simulated trials, we can quantify the sim-to-real gap in navigation tasks, providing insights into areas where the digital twin may need refinement.

Localization Accuracy. Localization accuracy is determined by comparing pose estimates from the *Google Cartographer* SLAM algorithm with ground truth data from the motion capture system (real-world experiments) or the digital twin (digital twin experiments). The *Root Mean Square Error* (RMSE) is used as a standard measure to quantify localization accuracy, capturing the average magnitude of deviations between the estimated and ground truth positions. The RMSE is given by Eq. 1. x_i^{est}, y_i^{est} represent the estimated position coordinates. x_i^{true}, y_i^{true} denote the ground truth position coordinates.

$$RMSE_{position} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left((x_i^{\text{est}} - x_i^{\text{true}})^2 + (y_i^{\text{est}} - y_i^{\text{true}})^2\right)} \tag{1}$$

Goal Accuracy. The ability to reach the assigned destination is evaluated using the *Euclidean distance* between the final and goal poses, as defined in Eq. 2, where (x_{final}, y_{final}) and (x_{goal}, y_{goal}) denote the coordinates of the final and goal positions.

$$d_{pos} = \sqrt{\left((x_{final} - x_{goal})^2 + (y_{final} - y_{goal})^2\right)} \tag{2}$$

Orientation error θ_{error} is determined by calculating the angular difference between the robot's final orientation θ_{final} and goal orientation θ_{goal}, as per Eq. 3.

$$\theta_{\text{error}} = |\theta_{\text{final}} - \theta_{\text{goal}}| \tag{3}$$

Path Consistency. The Hausdorff distance is a commonly used metric for measuring similarity between two point sets [42]. In this study, it is applied to assess the consistency between the robot's real-world and digital twin navigation paths. The Hausdorff distance quantifies the maximum deviation between two sets of path points, capturing spatial discrepancies. The distances between pairs of points are calculated using the *Euclidean metric*. Here, A and B are sets of path points from the real-world and digital paths, respectively. $d(a, b)$ is the *Euclidean distance* between points $a \in A$ and $b \in B$. inf and sup represent the minimum and maximum operators. The Hausdorff distance is defined as:

$$H(A, B) = \max\left(\sup_{a\in A}\inf_{b\in B} d(a, b), \sup_{b\in B}\inf_{a\in A} d(a, b)\right) \tag{4}$$

Navigation Performance. Reliability during navigation with *Nav2* is assessed by monitoring the number of collisions and instances of human intervention required to resolve navigation failures. This custom-defined metric provides insights into the robot's ability to operate safely in cluttered or constrained environments. N presents the total number of navigation trials, C the number of collisions during the trials and I the number of instances where human intervention is required to resolve failures. The Reliability Score R, where higher values indicate better reliability, is defined as:

$$R = \frac{N - (C + I)}{N} \tag{5}$$

Navigation Uncertainty Using Confidence Intervals. A three-step methodology quantifies the uncertainty of navigation predictions in the digital twin. For each scenario, digital twin paths are aligned to a reference path, calculated as the mean path of the simulations, by identifying and selecting the closest points from each path to the reference. The *Euclidean distances* between the aligned path points and the points of the reference path are calculated to quantify deviations. Bootstrapping is employed to estimate 95 % confidence intervals for these distances, capturing variability. The maximum upper bound of all confidence intervals serves as an indicator of a conservative estimate of the deviation between digital twin paths. For each experiment, the digital twin and real-world path are aligned and deviations are quantified as the *Euclidean distance* between

the two sets of paths. Bootstrapping is applied to compute 95 % confidence intervals. From all experiments, the maximum upper bound of all confidence intervals is extracted to represent the conservative estimate of the deviation between digital twin and real-world paths.

The final confidence interval is computed as the sum of the confidence interval widths from the digital twin paths $CI_{digital}$ and the digital-twin-to-real comparison $CI_{digital-to-real}$:

$$CI_{total} = CI_{digital} + CI_{digital-to-real} \tag{6}$$

CI_{total} quantifies the uncertainty in the navigation path of a digital robot, providing a worst-case estimate of how much a real-world path may deviate from the digital twin outcome.

By employing the presented method, we can conduct a detailed comparison of the navigation behavior between the real robot and its digital counterpart, enabling the assessment of the digital twin's validity. This approach facilitates the identification of discrepancies, thereby supporting its application in industrial automation and flexible assembly systems.

5 Results

In the following, we present a quantitative analysis of the robot's navigation behavior to evaluate the alignment between real-world and digital twin performance. The research data and evaluation code are available on Zenodo [43].

5.1 Quantitative analysis

Localization accuracy. Localization accuracy is assessed by comparing the pose estimates from the SLAM algorithm to ground truth data from motion capture and digital twin. The RMSE was calculated for each trial. Results showed consistent localization accuracy across all trials with comparable performance in real-world and digital twin. Figure 5 shows the results of all experiments with a mean RMSE in the digital twin of 0.033 m and 0.028 m in the real-world.

Goal accuracy. The robot's ability to reach its goal is evaluated by measuring the *Euclidean distance* between its final position and the goal position, alongside the orientation error between the final and desired orientations. Across all experiments, the mean positional error in digital twin was 0.07 m, with a maximum of 0.147 m, while in real-world experiments, the mean positional error was 0.055 m, with a maximum of 0.1 m. The mean orientation error in the digital twin was 4.157°, peaking at 5.889°, whereas

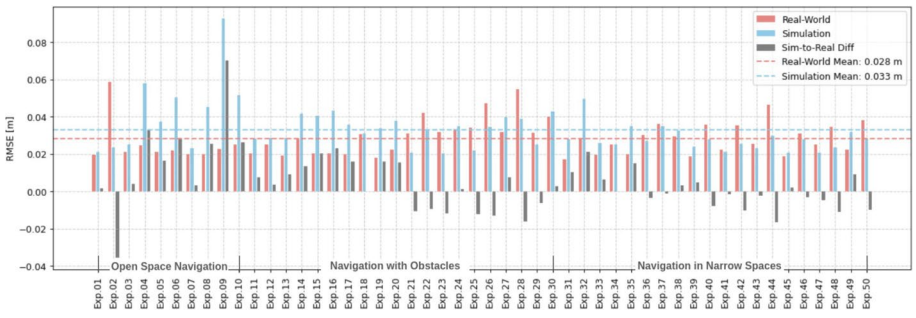


Fig. 5 Localization accuracy across all experiments in real-world and digital twin, evaluated using the RMSE metric

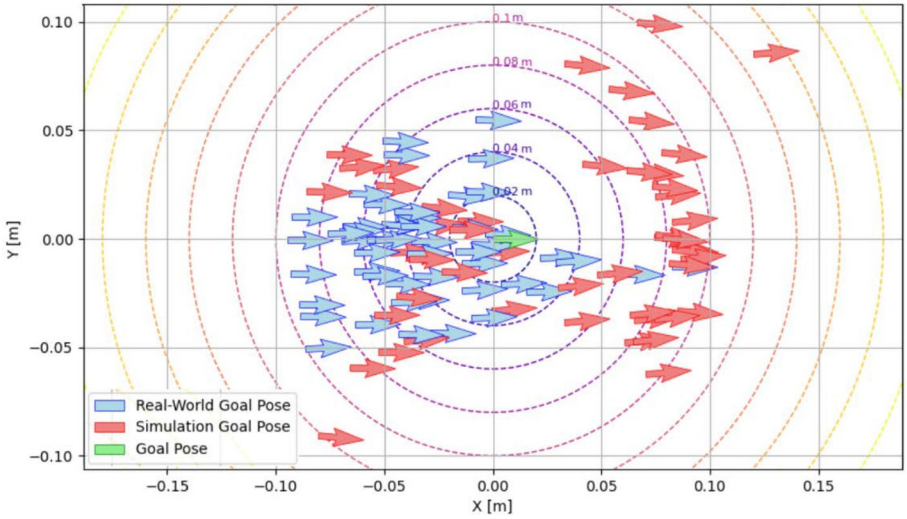


Fig. 6 Goal pose accuracy across all experiments in real-world and digital twin

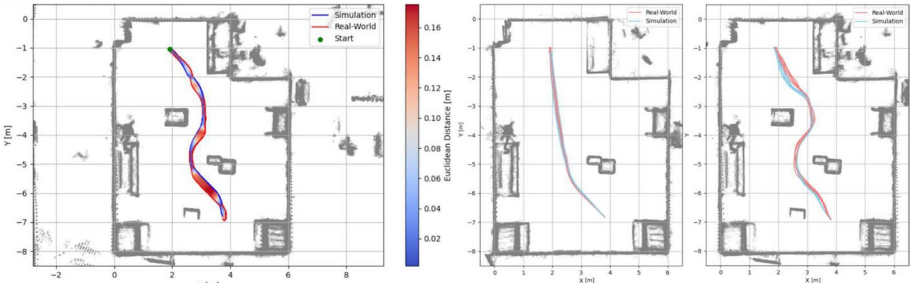


Fig. 7 All plots present a top-down view of the robot's environment, generated from point cloud recordings, with simulated paths shown in blue and real-world paths in red. The left plot visualizes the consistency between the simulated and real robot paths by showing the color-coded Euclidean distance between them. The middle and right plot visualize the paths of all conducted experiments for specific scenarios. Path consistency for experiment in scenario 05: 'Navigation in Narrow Spaces, Known Obstacles' (left). Comparison of all navigation experiments for scenario 02: 'Navigation with Obstacles, Unknown Obstacles' (middle) and scenario 04: 'Navigation in Narrow Spaces, Unknown Obstacles' (right)

in real-world experiments, it was 1.473°, with a maximum of 3.949°. Figure 6 shows the results across all experiments.

Path consistency. Path consistency is quantified using the Hausdorff distance. Figure 7 compares the results of all navigation experiments for scenario 02 (*Navigation with Obstacles, Unknown Obstacles*) and scenario 04 (*Navigation in Narrow Spaces, Unknown Obstacles*), alongside a scenario 05 experiment (*Navigation in Narrow Spaces, Known Obstacles*), showing the real-world robot's and its digital counterpart's Navigation behavior, resulting in a Hausdorff distance of 0.175 m. Among all scenarios, scenario 05 (*Navigation in Narrow Spaces, Known Obstacles*) exhibited the lowest mean Hausdorff distance at 0.135 m, while scenario 01 (*Open Space Navigation*) demonstrated the highest mean Hausdorff distance at 0.25 m.

Navigation Performance. The robot's navigation performance is evaluated by tracking the number of collisions and instances requiring human intervention. Across all conducted experiments, including those in cluttered environments and scenarios with unknown obstacles, the system exhibited great reliability, with no collisions or

interventions recorded. This resulted in a reliability score of $R = 1$ for both real-world and digital twin experiments.

Navigation Uncertainty Using Confidence Intervals. The proposed method quantifies the uncertainty of navigation predictions in the digital twin using confidence intervals to evaluate the transferability of the digital twin behavior to real navigation processes. The confidence interval $CI_{digital}$ is 0.11 m, reflecting the consistency of paths in the digital twin for experiments with identical start and goal poses. The confidence interval $CI_{digital-to-real}$ for comparing digital twin and real-world experiment outcomes is 0.118 m. Combining these, the final confidence interval CI_{final} is 0.229 m, representing a conservative estimate of the maximum uncertainty in real-world navigation outcomes derived from digital twin predictions. Figure 8 shows a visual representation of the confidence intervals for navigation uncertainties, derived from the digital twin and real-world paths across three scenarios.

To provide a comprehensive overview, the results of all experiments are summarized in Table 1.

5.2 Qualitative observations

During the real-world experiments, the robot demonstrated reliable navigation performance across all scenarios. Its ability to localize and align with the goal pose was consistent. In scenarios involving unknown obstacles, the robot showed adaptive behavior, dynamically adjusting its trajectory to avoid collisions while maintaining progress toward the goal. In scenarios with known obstacles, the robot relied heavily on the premapped information, resulting in minor deviations between the original plan and the driven path. Conversely, when navigating unknown obstacle scenarios, the robot relied mostly on real-time sensor data, occasionally exhibiting minor oscillations when navigating the area. Despite these challenges, the robot consistently reached its goal pose without collisions or requiring external intervention. For all experiments, the robot was able to meet the configured goal completion criteria.

In the digital twin, the robot showed navigation behavior closely resembling that of its real-world counterpart. In both known and unknown obstacle conditions, the digital robot mirrored the real robot's path-planning, path-following, and obstacle avoidance

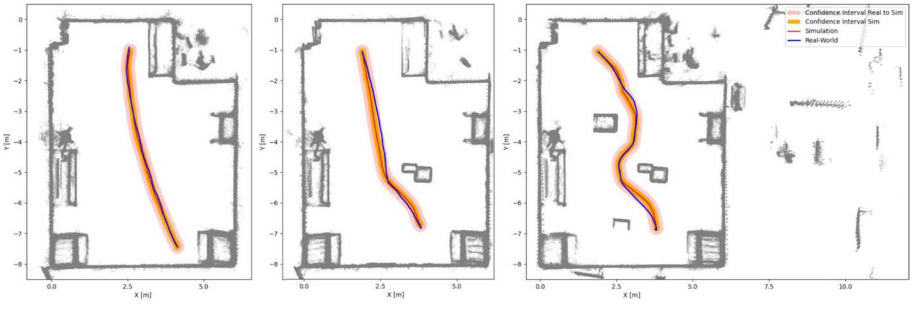


Fig. 8 The plots present the navigation uncertainty analysis for digital twin and real-world paths across three scenarios, with simulated paths shown in blue and real-world paths in red: "Open Space Navigation 01" (left), "Navigation with Obstacles, Unknown Obstacles 03" (middle), "Navigation in Narrow Spaces, Known Obstacles 05" (right). The yellow area illustrates the confidence interval $CI_{digital}$ of 0.11 m, reflecting the consistency of paths in the digital twin. The light red area illustrates the confidence interval $CI_{digital-to-real}$ of 0.118 m, reflecting the consistency between the digital twin and real-world experiment outcomes. The final confidence interval CI_{final} of 0.229 m is depicted by the entire colored area. It represents the region where the real robot is most likely to operate and serves as a predictive estimate of its movement

Table 1 Quantitative results of navigation experiments

Scenario	Obstacles known?	Type	Localization Accuracy RMSE [m]			Goal Accuracy Translation Error [m]			Goal Accuracy Orientation Error [°]			Path Consistency Hausdorff distance [m]			Reliability Score
			Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	
Open Space Navigation 01	–	Simulation	0.043	0.093	0.021	0.049	0.122	0.008	4.223	5.889	1.897	0.250	0.342	0.091	1
		Real-World	0.025	0.059	0.020	0.059	0.100	0.024	2.220	3.727	0.802				1
Navigation with Obstacles 02	No	Simulation	0.035	0.043	0.028	0.075	0.121	0.038	4.960	5.424	4.303	0.183	0.287	0.110	1
		Real-World	0.022	0.031	0.018	0.058	0.095	0.023	1.045	2.793	0.387				1
Navigation with Obstacles 03	Yes	Simulation	0.031	0.043	0.020	0.077	0.089	0.063	5.211	5.500	4.781	0.135	0.183	0.088	1
		Real-World	0.038	0.055	0.031	0.044	0.067	0.014	1.549	2.304	1.078				1
Navigation in Narrow Spaces 04	No	Simulation	0.031	0.049	0.024	0.088	0.147	0.052	5.082	5.552	4.280	0.197	0.246	0.145	1
		Real-World	0.026	0.036	0.017	0.055	0.094	0.023	1.372	3.949	0.083				1
Navigation in Narrow Spaces 05	Yes	Simulation	0.025	0.032	0.021	0.061	0.090	0.024	1.309	3.669	0.209	0.209	0.288	0.128	1
		Real-World	0.030	0.046	0.019	0.061	0.077	0.028	1.177	2.116	0.174				1

strategies. The robot made identical decisions in both the digital twin and real-world environments, such as choosing the direction to navigate around obstacles (left or right), indicating a high degree of behavioral consistency across both domains. As in the real-world, the robot consistently reached its goal pose without collisions or external intervention. However, in 3 experiments, the digital robot slightly overshot the goal, likely due to the joint parameters of the mecanum wheels, suggesting the need for further fine-tuning. Consequently, the robot did not meet the goal completion criteria in 3 out of 50 experiments.

Overall, quantitative and qualitative comparison between real-world and digital twin experiments revealed consistency in the robot's navigation behavior. Nevertheless, subtle differences in path execution were observed, underscoring the impact of digital twin fidelity — both in terms of the robot model and the environmental model — on navigation behavior. Despite these differences, the digital twin proved valuable for providing insights into the likely path the real robot might take, making it a useful tool for preliminary analysis and planning of robotic tasks in flexible assembly systems. Importantly, these observations underscore the value of the proposed method itself: the systematic validation approach enables detailed comparisons between simulation and reality, providing a structured way to assess behavioral fidelity. This comparison would not be possible without such a method, reinforcing its relevance for evaluating digital twins in industrial robotics.

6 Conclusion and future work

This paper presented a systematic methodology for validating the predictive fidelity of physics-based digital twins in robotic Navigation tasks, bridging a critical gap in sim-to-real transferability for industrial mobile manipulators. By systematically comparing 50 real-world and digital twin trials in a flexible assembly system, we derived a confidence interval to quantify navigation uncertainty and assess the transferability of the digital twin's behavior to real-world navigation. The results demonstrated strong alignment between digital twin and real-world behavior across all scenarios, with a navigation uncertainty of ±0.229 m at 95% confidence. The analyzed metrics revealed consistent performance, with minor deviations attributed to factors like the robot model's physics parameters and environmental fidelity (mean Hausdorff distance of 0.195 m and localization RMSE differences of 0.005 m). Furthermore, qualitative observations underscored the digital twin's ability to replicate key aspects of the robot's navigation behavior, such as trajectory patterns and decision-making around obstacles. However, discrepancies in goal achievement, such as slight overshooting, highlighted the need for further fine-tuning of the digital twin physics. While the digital twin may not perfectly replicate all real-world scenarios, it offers a reliable tool for assessing navigation strategies before deployment in physical environments.

Future efforts will focus on addressing the identified limitations and expanding the digital twin's applicability.

- *Improved Robot Modeling:* Enhancing the fidelity of the digital robot's physics through additional experiments comparing the responses of the real and digital robots to identical control commands. This addresses observed deviations in goal accuracy and is expected to reduce overshooting in the digital twin experiments.

- *Environment Fidelity:* Increasing the level of detail in environmental models, including CAD-based object representations, to better replicate real-world complexity. This is expected to result in digital point cloud measurements that more closely match those from the real environment, leading to improved localization and navigation behavior alignment between digital twin and reality.
- *Real-Time Adaptation:* Exploring the integration of real-time feedback from robot sensors, perception modules, and the OptiTrack system to dynamically update the digital twin, ensuring it reflects changes in the real environment. This targets the current limitation of static digital environments, which may not account for unmodeled or dynamic real-world conditions.
- *Validation Pipeline:* Integrating an automated pipeline to test and validate robotic navigation behavior in the digital twin against safety and performance thresholds prior to real-world execution. This addresses the current limitation of offline evaluation and supports scalability. Moreover, it establishes a foundation for transitioning to a fully integrated digital twin with real-time validation capabilities.
- *Method Refinement:* Incorporating uncertainties from individual system components, such as sensor noise, actuation inaccuracies, and environmental variability, into the validation method. This extends beyond the current approach, which focuses primarily on path deviation without isolating the impact of specific sources of error.

In conclusion, the findings demonstrate the predictive fidelity of physics-based digital twins in robotic navigation tasks, providing a foundation for their use as reliable forecasting tools. This supports greater industry confidence in leveraging digital twins for planning and validating mobile robot autonomy in flexible and dynamic assembly environments.

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Author Contributions
L.B. implemented the system, developed the methodology, conducted the experiments, and wrote the main manuscript text. M.H., A.M., F.K.S.C., Y.N.L., and X.P. contributed to the conceptualization of the study and reviewed the manuscript. A.G., B.D., and Y.N.L. provided supervision and manuscript review. R.S. supervised the overall research project. All authors reviewed and approved the final manuscript.

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Data Availability
The datasets collected from the real-world and digital twin, along with all results and scripts used to evaluate the data, are available for open-source access. The data and resources are hosted on ***Zenodo*** [43].

Code Availability
The resources used in this study are owned by the Laboratory for Machine Tools and Production Engineering (WZL) at RWTH Aachen University. Access to these materials is available upon request through the corresponding author, subject to institutional approval and availability. The code developed for the evaluation is hosted on ***Zenodo*** [43].

Declarations

Ethics approval and consent to participate
All participants gave informed consent to participate in this study.

Consent for publication
All authors have consented to the publication of this manuscript.

Consent to participate
Not applicable. This study did not involve human participants.

Clinical trial number
Not applicable.

Conflict of interest
The authors have no Conflict of interest to declare that are relevant to the content of this article.

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